

Uncertified products

Taiwanese companies not getting their USB 3.0 products certified

Gamechanger?

Adieu HDMI, DVI, Ethernet, USB, eSATA, Firewire. We'll explore this possibility in the months to come

From the labs



A laptop running a full HD stream using Light Peak technology

An important point to note would be that optical fibre cables themselves have nearly infinite bandwidth, so the upgrade potential is huge and Intel is already promising speeds of 100 Gbps over the next decade.

Light Peak consists of a controller chip and an optical module that converts an electric signal (primarily the information you wish to transfer) into light. Even the reverse is performed. This leaves the remaining circuitry unchanged. With mass sale and production, the economies of scale will be leveraged and the cost of each module is expected to drop significantly. Given the relatively high short distance communication needs compared to under sea cables used in the telecom industry, the detector and the laser (the signal source) needn't be of extremely high quality. This helps in reducing costs.

The primary motivation to move to an optical medium from an electrical medium comes to the fact that the limits for speed, length and issues regarding electromagnetic interference (EMI) are proving inhibitors in any major upgrade to transmission speeds now. Optical communication technology isn't in itself new

and have been used in data centres and telecommunications for a long time, Light Peak brings similar capabilities to consumer electronic devices at a reasonable price.

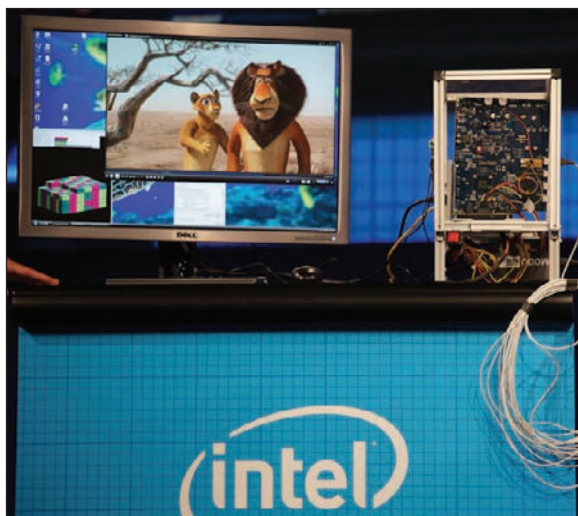
Intel is already working in full steam with component manufactures to facilitate commercial deployability as early as the end of 2010. The aim is to replace all copper wire by silicon photonic systems at a later and Light Peak is a start in that direction. Sony and Apple seem to be on the forefront along with intel for Light Peak implementation. As a matter

of fact, the first public demonstration of the Light Peak technology was made on a Hackintosh. Once a complete conversion to optical transmission is done with all components primarily communicating with each other via photons instead of electrons, it will further provide a much fillip to communication speeds as there will be no redundant signal conversions.

There's speculation also that there's more to Light Peak. It has been suggested by several insiders that Apple initially brought the concept to Intel to build it, as early as 2007. Their plans included an inter operable standard that could handle massive amounts of data and could do away with the connector mess by only using a single connector type. We are already aware of Apple's love for minimalistic design. So don't be surprised if an Apple Mac supports Light Peak slightly before your PC.

Even for Light Peak, currently it is not a glorious road. There are major concerns in several departments, especially power efficiency due to the constant need of photo generation. A declarative stand is yet to be released by Intel on this issue. Another major concern is the ability of connected devices to store and process data at such speeds. For example, your hard disk may not be able to write data as fast as it receives from Light Peak cables, though of course you are in luck if you are dealing only with solid state storage devices.

Having looked at the near future of device communications, we would wind up by informing you about another rotary initiative by Intel called 'Hybrid Silicon Laser' technology. It uses four channel. Laboratory prototypes demonstrate speeds of 50 Gbps already, but given the massive amount of effort gone in manufacture synchronisation, media publicity and marketing, don't expect it to be commercially available anytime soon. Nevertheless, breakthroughs over here can be implemented in the more glamorous sister going by the name of 'Light Peak'. **d**



Notice the magic glass-box on the right of the picture